Why the words float

The working behind every number on the approved sheet of 2026-11-01: the mirror in the glass, the backwards text, cube against sheet, where the light goes, how big the words are, what the display and the radio ask of one small regulator, a resistor for a filament, and four legs under 314 g of glass.
Read after Desk Hologram Build. Nothing measured yet; every figure is derived from datasheet numbers and the sheet's dimensions.

The trick in one sentence

Half of the light from every pixel bounces off a mirror hidden inside a block of glass and comes out of the front face, so your eye traces it back to a picture standing upright at the back of the block, 33 mm in, where nothing is.

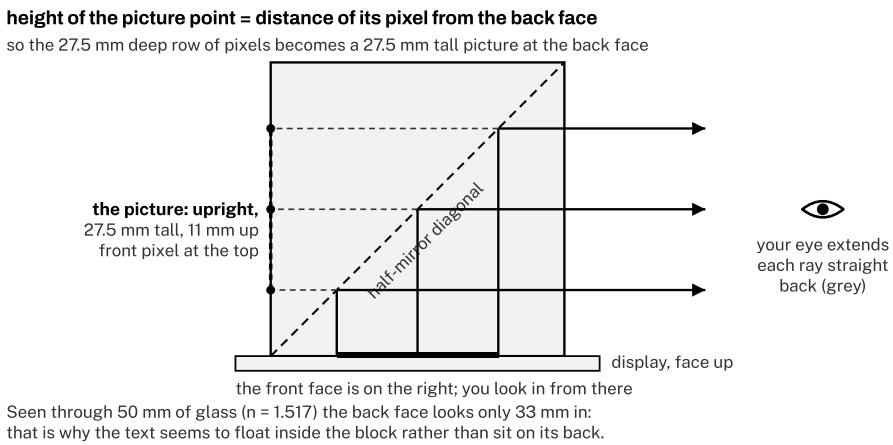
Everything on the approved sheet follows from that one sentence: why an OLED, why a cube and not a sheet, why 50 mm, why the text is written backwards, why the room behind the cube should be dark. This page takes each decision and shows the working, in the order the sheet made them.

Where the picture stands

A beam-splitter cube is two right-angle prisms glued face to face, with a coating on the glued face that reflects half the light and passes half. Sitting on the display, that coating is a 45° mirror across the cube, from the bottom edge at the back to the top edge at the front.

A pixel sends its light straight up. At the diagonal half of it turns 90° toward you. Your eye does not know about the turn: it extends the ray straight back, and the extension lands on the back face of the cube, as high above the base as the pixel was in front of the back face. Height equals depth, one for one.

So the 27.5 mm deep strip of pixels becomes a 27.5 mm tall picture, standing on the back face, with the display's front edge at the top and the back edge at the bottom. The picture is the same size as the display, not magnified, and it is a virtual image: no light actually comes from where you see it, which is the whole charm.



The construction. Three pixels, three turns at the diagonal, three grey extensions meeting the back face. The picture occupies 11 to 39 mm above the cube's base because the pixels sit centred under it.

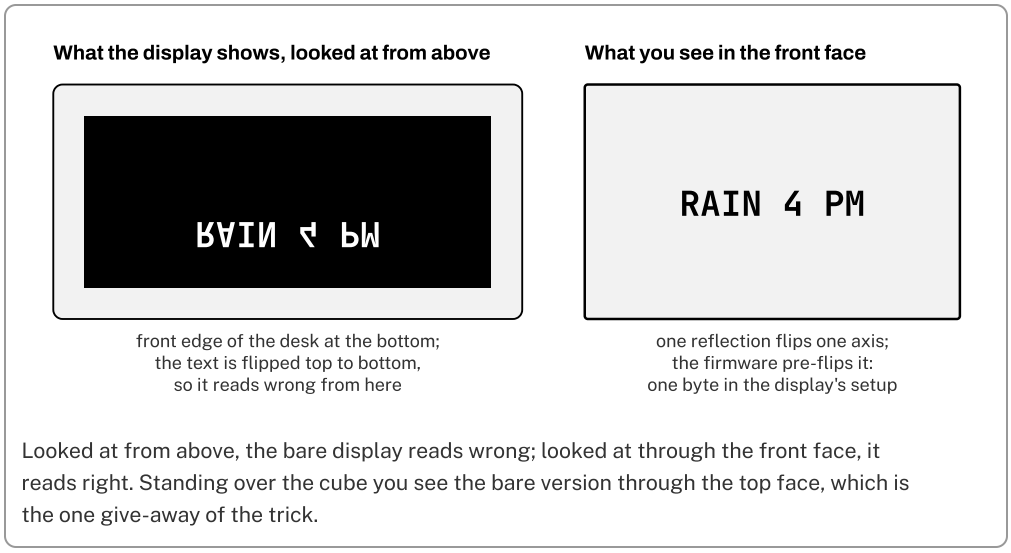
One more effect makes it look like it floats inside the glass rather than painted on the back. Light leaving glass bends away from the normal, so anything seen through a slab appears closer than it is, by the index of refraction: 50 mm of K9 glass (n

= 1.517) looks 33 mm deep. The text seems to hang two thirds of the way in, and when you move your head it slides against the back face exactly as a thing at that depth would.

Why the text is written backwards

A single mirror flips one axis. Look at the construction again: the pixel nearest you ends up at the top of the picture, the farthest at the bottom, while left and right stay where they are. That is a top-to-bottom flip relative to the display, which is the same thing as a left-to-right flip of the letters as you would read them in the cube.

The fix is free. The display's controller has a one-byte command that reverses the order it maps columns to memory (segment remap) and another that reverses the row scan. The firmware sets the right one during setup, draws normally, and the mirror undoes the pre-flip. Which byte it is depends on which way round the module is mounted, so the sheet says: power it, read it through the cube, and swap the other bit if it is upside down instead of mirrored.



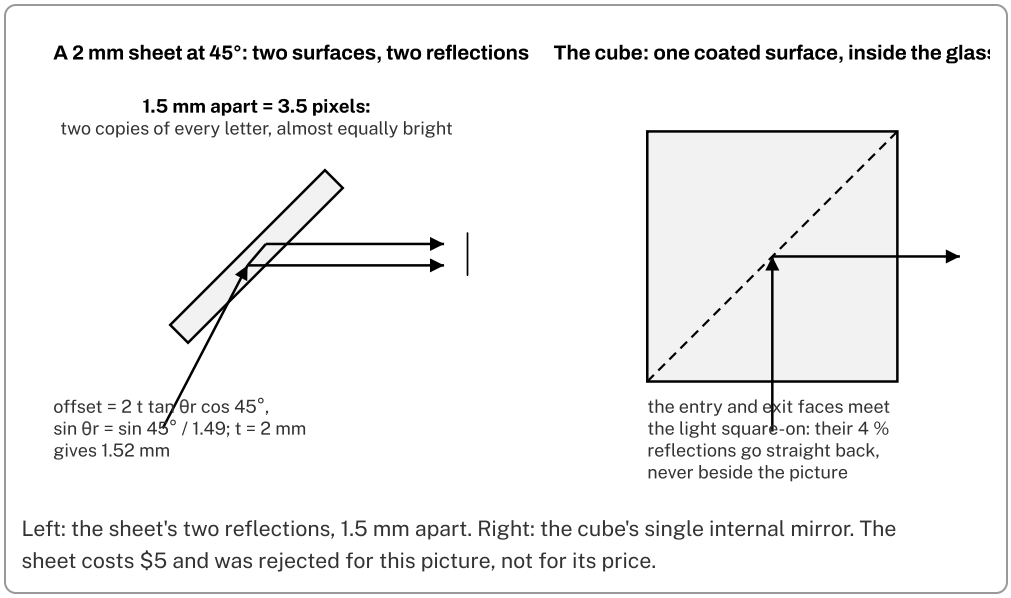
Looked at from above, the bare display reads wrong; looked at through the front face, it reads right. Standing over the cube you see the bare version through the top face, which is the one give-away of the trick.

Why a cube and not a sheet

The cheap version of this effect is a sheet of clear acrylic at 45°. It fails on the fine detail. A sheet has two surfaces and both reflect about 4 %: the front one directly, the back one after the light has gone in, crossed the thickness and come out again. The two reflections leave parallel but displaced, by 2 t tan θr cos 45°, where θr is the refracted angle inside the plastic.

For a 2 mm sheet that is 1.5 mm, and at 0.43 mm per pixel that is 3.5 pixels: every stroke of every letter appears twice, side by side, almost equally bright. Small text turns to fuzz. A coated plate helps, but the uncoated side still adds a faint second copy.

In the cube the reflecting coating is inside the glass, at the glued diagonal, and the faces the light crosses are square to it. Their 4 % reflections go straight back the way they came and never land next to the picture. One coating, one copy. The cube also needs no frame to hold it at 45°, no black box, and at 314 g it does not move: three decisions solved by one part.

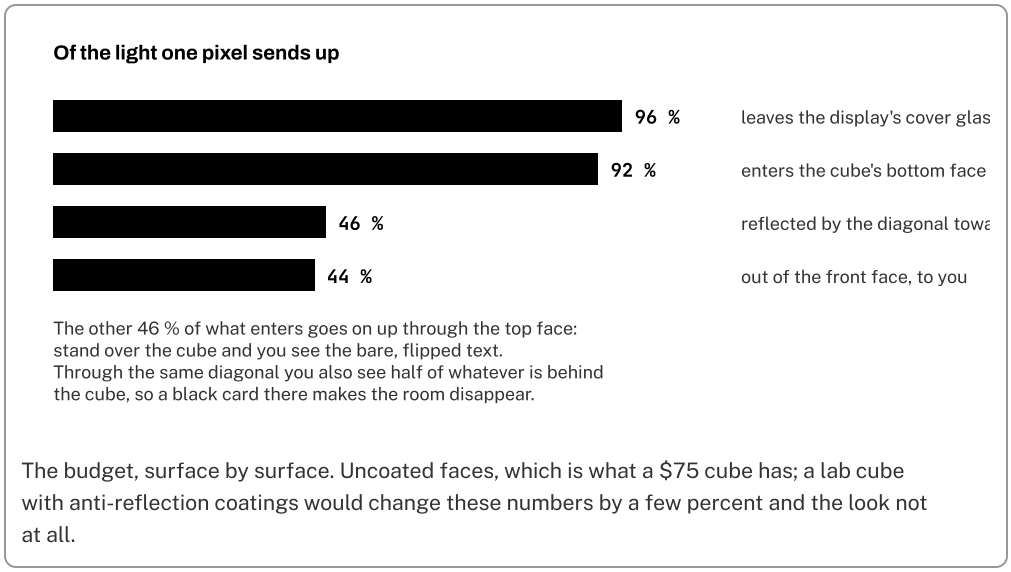


Left: the sheet's two reflections, 1.5 mm apart. Right: the cube's single internal mirror. The sheet costs $5 and was rejected for this picture, not for its price.

Where the light goes

Each air-to-glass surface reflects $((n - 1)/(n + 1))^2$ of the light, 4.2 % for this glass. A pixel's light crosses the display's own cover glass, an air gap, the cube's bottom face, the 50 % diagonal and the front face: about 44 % reaches you. The rest mostly carries on up through the top face.

That is bright enough. The display is built to be read in daylight at full power and here it only has to be read in a room, through glass, at arm's length. It also means the picture is only as dark as what is behind it: through the same 50 % diagonal you see half of the room behind the cube. A black card standing behind the back face turns that half into nothing, and the words hang on black.



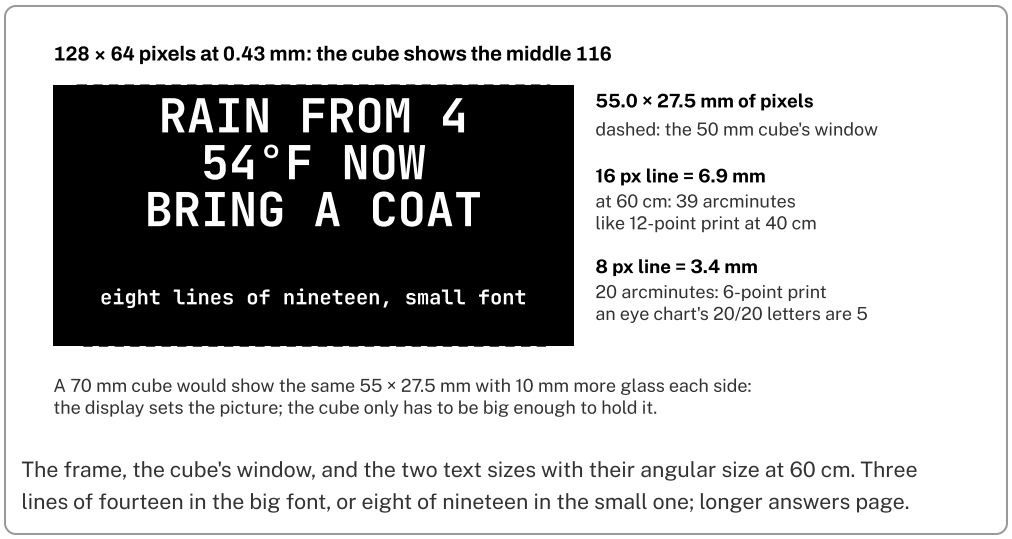
The budget, surface by surface. Uncoated faces, which is what a $75 cube has; a lab cube with anti-reflection coatings would change these numbers by a few percent and the look not at all.

How big the words are

The display has 128 × 64 pixels at 0.43 mm, 55 × 27.5 mm. A 50 mm cube covers 50 of the 55 mm, so the firmware uses the middle 116 columns and the picture inside the glass is 50 × 27.5 mm.

A 16-pixel line of text is 6.9 mm tall. Seen from 60 cm, a seated viewer's distance, that subtends 39 arcminutes, about what 12-point print subtends on a page held at 40 cm. The small 8-pixel font is 3.4 mm and 20 arcminutes, like 6-point print: readable, not comfortable. An eye-chart letter you can just read is 5 arcminutes, so both fonts have room to spare.

That is why 50 mm. A 70 mm cube would hold the same 55 × 27.5 mm of pixels with 10 mm of empty glass each side: more object, no more picture, and 861 g. Bigger words need more pixels, which means a bigger display, which is a different build. The builder looked at 70 mm on 2026-11-01 and kept 50.



The frame, the cube's window, and the two text sizes with their angular size at 60 cm. Three lines of fourteen in the big font, or eight of nineteen in the small one; longer answers page.

Why an OLED, and what it asks of the 3.3 V rail

An OLED pixel is a light source. A black pixel is off and sends nothing into the cube, so black on the display is truly empty in the picture. An LCD cannot do that: its backlight shines through black pixels too, and the cube would show a grey rectangle with the words in it. The beam splitter only works with an emissive display, and the monochrome white one is the brightest per watt the builder owns.

Being a light source, the display's current follows the number of lit pixels. Every pixel on draws 50 to 75 mA; a few words on black, about a tenth of the pixels, draw about 10 mA. The pixels themselves want 12 to 13 V, which the module makes from 3.3 V with an on-board charge pump. That pump takes its current in gulps, which is why the kit ships a 220 µF capacitor and the sheet puts it at the header pins, not 40 mm away at the controller.

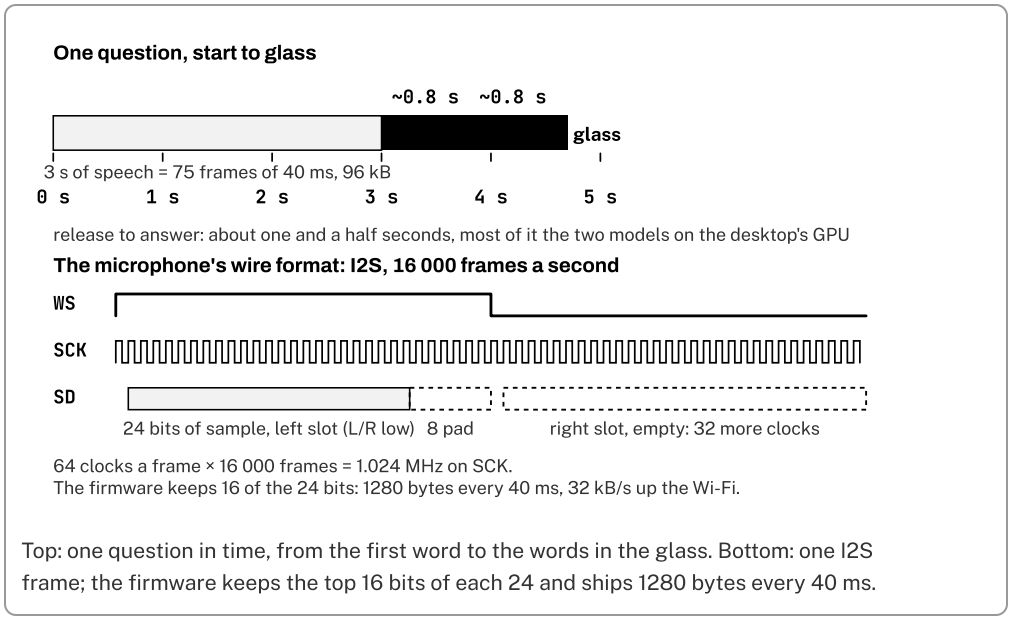
The picture is 1024 bytes, one bit per pixel. At the display's rated 4 MHz on SPI that is 2.0 ms per frame; the hologram redraws a dozen times a second at most, so the bus is idle almost always. The module speaks SPI only when two solder jumpers on its back say so (BS1 = 0, BS2 = 0), which is why the sheet asks for a photograph of the back before the header goes on.

The microphone and the sixteen thousand

The microphone is a MEMS capsule: a silicon diaphragm over a fixed plate forms a capacitor whose value wobbles with air pressure, and a converter on the same chip turns that into a 24-bit number 16 000 times a second. It leaves the capsule already digital, on the three I2S wires, so nothing analogue is wired and nothing picks up the radio's hum.

Sixteen thousand samples a second captures everything up to 8 kHz, which covers speech: the consonants that tell words apart live between 2 and 6 kHz. Whisper itself resamples everything it hears to 16 kHz, so sending more would be thrown away. On the wire each frame is two 32-bit slots, left and right; with L/R tied to ground the capsule answers in the left one, and the clock runs at 1.024 MHz.

The capsule hangs under the table top with its port facing the desk. That is fine for sound. A 72 mm board is small next to the wavelengths of speech, 34 cm at 1 kHz, so sound bends round it; only the highest sibilants, above 5 kHz, lose a little. The desk reflects some sound back into the port, which Whisper has heard a million times before.



Top: one question in time, from the first word to the words in the glass. Bottom: one I2S frame; the firmware keeps the top 16 bits of each 24 and ships 1280 bytes every 40 ms.

From breath to words to glass

While the button is down the controller packs 40 ms of samples, 1280 bytes, into one binary WebSocket frame and sends it: 32 kB a second, a thirtieth of what the Wi-Fi link carries. The server buffers the frames; on release it hands the whole utterance to Whisper, hands the text to the model, and sends the reply back as one JSON message. Nothing is streamed back but text, because this device has no speaker.

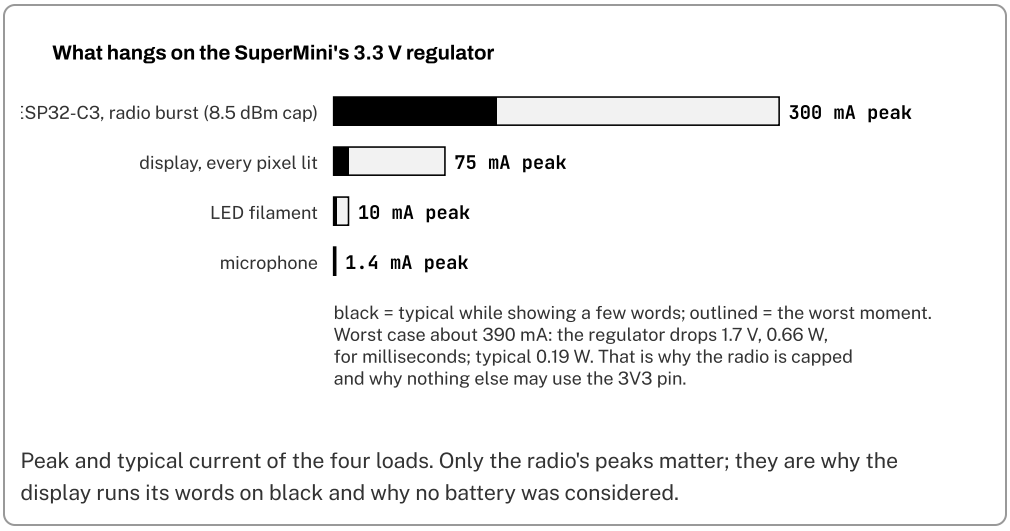
The budget from release to glass is about a second and a half. Whisper small.en on the desktop's GPU transcribes a 4 s question in about 1.1 s; gpt-oss:20b at reasoning effort low answers a short question in half a second to a second once warm. The character prompt asks for two sentences because the glass holds about forty words a page.

Push to talk instead of a wake word is a C3 decision. A wake-word model needs a megabyte or more of RAM to run always-on; the C3 has 400 kB and no external memory, and an always-on microphone on the desk is a different kind of object anyway. The button also makes the start and end of the utterance exact, which is the part Whisper is worst at guessing.

One 3.3 V regulator

USB brings 5 V to the SuperMini, and a small low-dropout regulator on the board makes the 3.3 V that everything runs on: the C3, the display, the microphone and the filament. The regulator is the busiest part of the build. Its worst moment is a radio burst with every pixel lit: about 390 mA, dropping 1.7 V, 0.66 W of heat in a part the size of a grain of rice. It survives because the burst lasts milliseconds; the typical draw with a few words showing is a third of that, 0.19 W.

Two sheet rules come from this. The radio is capped at 8.5 dBm, as on the first Pocket AI, which halves its peak current and still reaches a router in the same flat. And nothing else may hang on the 3V3 pin: the display, the capsule and 10 mA of filament are the whole list.



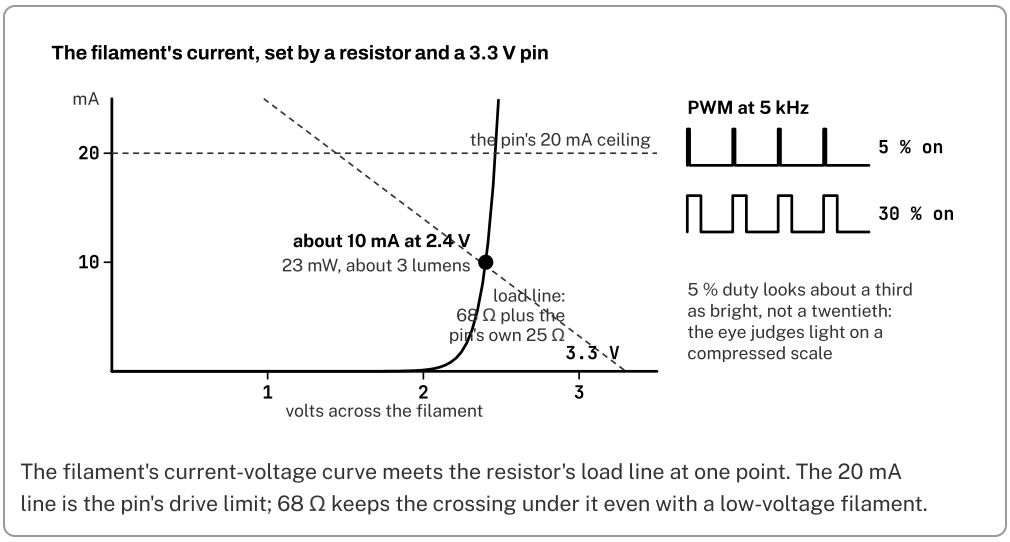
Peak and typical current of the four loads. Only the radio's peaks matter; they are why the display runs its words on black and why no battery was considered.

The amber filament

An LED filament is a row of tiny blue LED chips on a ceramic strip under a phosphor coating; the thicker and more orange the phosphor, the warmer the light. 2200 K is candle-coloured. Rated at 3 V and 2 W it would pour out 200 lumens; this build wants a glow, not a lamp.

An LED does not take a voltage and decide a current. Its current climbs exponentially with voltage, so 0.1 V too much is a lot of current and a dead filament. A resistor fixes that: the pin gives 3.3 V, the resistor and the pin's own 25 Ω take whatever the filament does not, and the two curves cross at one point, about 10 mA at 2.4 V (the sheet's 12 mA counts the resistor alone). That is 23 mW and roughly 3 lumens, a quarter of a candle, a soft pool under the table top in a dim room and invisible in daylight, which is what was asked for.

The same pin dims it by pulse-width modulation: on for 5 % of each 200 µs at rest, 30 % while listening, breathing while thinking. The eye compresses brightness roughly as a cube root, so 5 % duty looks like a third, not a twentieth; the firmware's levels are set by eye, not by arithmetic. If a real lamp is ever wanted, the owned MOSFET and the 5 V pin give 200 mA in one more part.



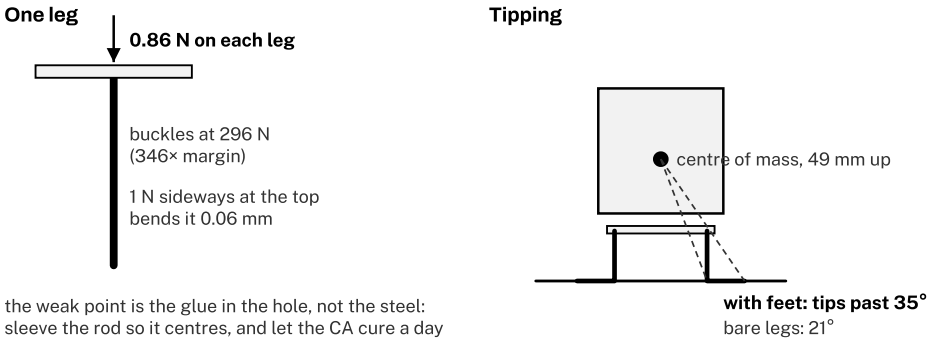
The filament's current-voltage curve meets the resistor's load line at one point. The 20 mA line is the pin's drive limit; 68 Ω keeps the crossing under it even with a low-voltage filament.

Four legs

The structure is four 1.5 mm stainless rods through the display module's own mounting holes. The whole device weighs about 349 g, 0.86 N on each leg. A 20 mm length of 1.5 mm rod fixed at the top buckles at about 296 N by Euler's formula, 346 times more; a 1 N sideways push at the top bends it 0.06 mm. The steel is not the limit. The glue in the hole is, which is why the rod is sleeved to centre it and the CA gets a day to cure.

Tipping is the real question with 314 g of glass on top. The centre of mass sits 49 mm up. With the legs bent into feet that reach 15 mm out front and back, the device has to lean 35° before it goes over; on bare vertical legs only 21°. The feet cost no extra rod and no solder: one bend per leg.

Stainless rather than brass because the material rule says so for every build after Cyber Dawn, and because here it costs nothing: no joints are soldered, the rods are glued, and 304's stiffness is 60 % higher than brass's at the same diameter, which is where the 0.06 mm came from.



Left: one leg's load against what would buckle or bend it. Right: the tipping angle with and without the feet; the dashed lines run from the centre of mass to the two footprints.

The numbers on the sheet, and where they come from

NUMBER	WHAT IT IS
33 mm	how deep the picture looks through 50 mm of glass (50 / 1.517)
27.5 mm	picture height: the depth of the row of pixels, one for one
11 mm	picture bottom above the cube's base: (50 − 27.5) / 2
116 of 128	columns inside the cube: 50 / 0.43
1.5 mm	double-image offset a 2 mm acrylic sheet would give: 3.5 pixels
44 %	of a pixel's light that reaches you through four uncoated surfaces and the 50 % diagonal
39 arcmin	a 16-pixel line of text seen from 60 cm; 20/20 letters are 5
2.0 ms	one 1024-byte frame at 4 MHz SPI
1.024 MHz	I2S bit clock: 16 000 frames × 64 bits
1280 B / 40 ms	one audio frame up the Wi-Fi, 32 kB/s
0.66 W	worst-moment heat in the 3.3 V regulator: 1.7 V × 390 mA
10 mA	filament current: (3.3 − 2.4) / (68 + 25)
296 N	buckling load of one 20 mm leg, fixed-free, 1.5 mm 304
35°	lean before tipping, with feet; 21° without

Which part each idea lives in

- **the half-mirror diagonal and the virtual image:** the beam-splitter cube
- **emissive pixels, the charge pump, the mirror byte, 4 MHz SPI:** the 2.42-inch display
- **the 3.3 V regulator, the radio cap, push to talk:** the ESP32-C3 SuperMini
- **MEMS capsule, I2S frames, 16 kHz:** the I2S microphone
- **the diode curve, the resistor, PWM dimming:** the LED filament
- **Euler buckling, tipping, glue in a hole:** the 1.5 mm stainless rod

What comes next

The firmware port: the first Pocket AI's character firmware with the amplifier off, the display driver moved to SPI with the mirror byte, the lamp on GPIO0, and text pages for the transcript. Then the cube and the filament arrive, the display's back gets photographed, and the first words go into the glass.

Built by `docs/pages/gen/physics_page.py` from the dimensions in `build_page.py`; it also writes `docs/PHYSICS.md`. Constants: K9 glass n = 1.517, acrylic n = 1.49, 304 stainless E = 193 GPa, pixel pitch 0.43 mm, viewer at 60 cm. The display's current and SPI limit are Adafruit's figures; Whisper and Ollama timings are the first Pocket AI's measured ones on this desktop. Nothing here has been measured on the hologram itself.